

tpm-pivot.md

Mode File — The Reallocation Engine

Mode	tpm-pivot.md
Author	Rohan — MS Information Systems, Northeastern University
Domain	Product Manager on OPT pivoting to Technical Program Manager
Version	v1.0 June 2025

1. What This Mode Does — and When to Use It

The TPM Pivot mode is for product managers — specifically international students on OPT or CPT — who are targeting Technical Program Manager roles but face a structural information gap: the job market doesn't cleanly separate 'PM-adjacent TPM' roles (cross-functional coordination, roadmap execution, stakeholder alignment) from 'pure-infra TPM' roles (incident command, reliability engineering, fleet operations). Applying indiscriminately burns OPT time and damages sponsorship odds at companies that historically don't hire non-engineers into TPM tracks.

This mode combines cognitive-demand scoring (Cognitive Pivot layer), H-1B sponsorship history (80 Days layer), and ATS liveness signals (Job-Ops layer) into a single triage workflow that helps a PM decide — before writing a single word of a cover letter — whether a posted TPM role is actually reachable given their background, their visa timeline, and whether the company has ever sponsored someone in that occupational category.

Primary user: PM on OPT (12-month or STEM extension) with $\leq$ 18 months remaining, targeting TPM roles at funded startups or mid-size tech companies.
Trigger condition: You have > 20 TPM postings in your tracker and no clear filter criteria.

2. Engine Layers & Data Sources

Layer	Source / Script	What it answers	Status
80 Days to Stay	data/h1b_companies.csv data/form_d_filings.csv	Has this company sponsored SOC 15-1299 (software/IT mgmt) in the last 3 years? Is it funded post-2022?	EXISTS
Job-Ops	scripts/check_liveness.py scripts/detect_ats.py	Is the posting still live? What ATS is it on (Greenhouse/Lever/Ashby)? Is the JD >90 days old?	EXISTS
Cognitive Pivot	data/soc_cognitive_scores.csv (SOC 11-3021 / 15-1299)	Does this role's SOC code score high on verification & system judgment — the cognitive premium a PM already holds?	EXISTS
TPM Role Classifier	scripts/tpm_classifier.py [PROPOSED — see §3]	Is this posting PM-adjacent or pure-infra based on JD keyword signature?	PROPOSED

3. Proposed New Component — tpm_classifier.py

The engine currently has no way to distinguish a TPM role that requires hands-on infrastructure engineering (on-call rotations, SLO ownership, incident command) from one that is essentially a senior PM with technical vocabulary. This matters acutely for PMs: the first type has a hard skills gap that sponsorship data won't fix; the second is a viable stretch. No existing dataset resolves this — it requires reading the JD.

Proposed script logic (not yet implemented):

```
# scripts/tpm_classifier.py – PROPOSED

# Input: raw JD text (from ATS scraper output)

# Output: role_type in {PM_ADJACENT, INFRA_HEAVY, AMBIGUOUS}

# + confidence score 0–1

PM_ADJACENT_SIGNALS = ['roadmap', 'stakeholder', 'cross-functional',
                        'OKR', 'PRD', 'launch', 'GTM', 'prioritization']

INFRA_HEAVY_SIGNALS = ['on-call', 'SLO', 'incident command', 'runbook',
                        'GPU fleet', 'reliability', 'capacity planning',
                        'MTTR', 'RCA', 'toil reduction']

# Classification: keyword frequency ratio + optional LLM tiebreaker
# LLM used ONLY after keyword scoring, for AMBIGUOUS cases only
```

Honesty note: This script does NOT exist in the repo yet. The workflow below marks any step using it as [INFERRED] rather than [VERIFIED]. It would take ~1 day to implement. The keyword lists above are a design spec, not a working classifier.

4. Mode Workflow — Step by Step

Step 1 H-1B + Funding Triage

[VERIFIED]

```
python scripts/query_h1b.py --soc 15-1299 11-3021 --company
```

Cross-reference data/h1b_companies.csv and data/form_d_filings.csv. Filter: sponsored >= 1 time in SOC 15-1299 or 11-3021 in 2022-2024 AND Form D filing within 36 months. Drop companies that fail both checks.

Step 2 Liveness + ATS Check

[VERIFIED]

```
python scripts/check_liveness.py --url
```

Confirm the posting is still active via Playwright. Flag postings >90 days old. Detect ATS provider (Greenhouse/Lever/Ashby). Greenhouse and Lever have better scraping coverage — Workday postings should be noted as lower-confidence.

Step 3 Role Type Classification

[INFERRED]

```
python scripts/tpm_classifier.py --jd # PROPOSED
```

Run keyword classifier on raw JD. Output: PM_ADJACENT / INFRA_HEAVY / AMBIGUOUS. INFRA_HEAVY roles get flagged — not dropped — with a manual-review annotation. If AMBIGUOUS: run optional LLM summarization to surface the 3 clearest requirement signals.

Step 4 Cognitive Demand Score Lookup

[VERIFIED]

```
python scripts/score_role.py --soc 11-3021
```

Pull verification + system-judgment sub-scores from data/soc_cognitive_scores.csv for the role's SOC code. PMs carry transferable premium here — this score is evidence you can use in the interview, not just a filter.

– aggregate output –

Write one row per role to `tpm_pivot_log.csv` (template in §6). Flag: APPLY / STRETCH / SKIP based on combined H-1B + liveness + role-type signals. STRETCH = viable but missing 1 signal. SKIP = no sponsorship history OR INFRA_HEAVY + no eng background.

5. What This Mode Can and Cannot Verify

CAN VERIFY (data-grounded)	CANNOT VERIFY (inferred / manual)
H-1B sponsorship in specific SOC codes — direct from USCIS data	Whether a company's TPM hiring has shifted since last fiscal year
Form D funding recency — SEC EDGAR filings	Whether 'Technical Program Manager' at this company means PM or SRE
ATS liveness — Playwright scrape at time of run	Hiring manager's actual tolerance for non-engineering TPM backgrounds
BLS cognitive sub-scores for SOC 11-3021, 15-1299	Role type (PM-adjacent vs infra) — until <code>tpm_classifier.py</code> is built
Posting age — ATS timestamp if available	Whether the company is actively running a TPM search vs. evergreen posting

6. Output Format + RUN_LOG.md Template

`tpm_pivot_log.csv` — one row per evaluated role:

company	role_title	soc_code	h1b_sponsored	form_d_months	liveness	role_type	cog_score	decision
Together AI	TPM – GPU Infra	11-3021	Yes (2023)	14 mo	LIVE	INFRA_HEAVY*	0.81	STRETCH
Stripe	TPM – Payments Platform	11-3021	Yes (2022,2023)	N/A – public	LIVE	PM_ADJACENT	0.81	APPLY
Ghost Startup	Technical PM	11-3021	No record	48 mo	DEAD	—	—	SKIP

`RUN_LOG.md` entry template:

```
## tpm-pivot | 2025-06-02
Input: 24 TPM postings from LinkedIn + Greenhouse scrape
H-1B filter (SOC 15-1299 / 11-3021): 24 → 17 companies with sponsorship history
Liveness check: 17 → 13 postings confirmed live
Role classifier: [INFERRED – tpm_classifier.py not yet built, manual review used]
PM_ADJACENT: 7 INFRA_HEAVY: 4 AMBIGUOUS: 2
Cognitive score lookup: SOC 11-3021 → 0.81 verification index, 0.74 system judgment
Final triage: APPLY 7 | STRETCH 4 | SKIP 2
Time to run: ~25 min (manual role classification is the bottleneck)
Next: build tpm_classifier.py to eliminate manual step
```

7. Domain Justification

Who uses this mode. Me — a PM with a CS background, on OPT, graduating in August 2026, actively interviewing for TPM roles. But the mode generalises to any PM who has technical fluency without a pure engineering track record and is targeting roles that sit on the PM/engineering boundary.

The information asymmetry. The label 'Technical Program Manager' is doing an enormous amount of variance-hiding. At Stripe it means cross-functional execution and roadmap discipline — a great PM fit. At Together AI or Anyscale it

means GPU fleet operations, on-call SLO ownership, and MTTR accountability — a role where a PM without infra experience will struggle in the interview and on the job. Nothing in a job title, a company's career page, or even most JDs makes this distinction legible. You only find out when you're in a loop and the hiring manager asks you to walk through how you'd run an incident postmortem for a CUDA OOM failure.

Engine layer connections. The 80 Days layer provides the visa-timeline filter that makes every other signal actionable — a PM_ADJACENT role at a company with zero sponsorship history is still a dead end. The Job-Ops layer catches dead postings before any effort is invested. The Cognitive Pivot layer is the least obvious connection but arguably the most useful for positioning: SOC 11-3021 (Computer and Information Systems Managers) scores 0.81 on verification tasks and 0.74 on system judgment in the BLS/O*NET data — exactly the cognitive profile a product manager builds. That score is evidence of role fit, not just a filter.

Failure modes.

Failure Mode	Who gets hurt worst	Why it's hard to catch
Role classifier marks a role INFRA_HEAVY based on JD language that is aspirational, not descriptive — the team is actually doing coordination work.	PM with strong instincts who self-selects out of a reachable role based on a false signal.	JDs are written by recruiters, not hiring managers. Infra vocabulary is often copied from a prior req. Only a recruiter screen or Glassdoor review would surface this.
H-1B history in SOC 15-1299 exists but the company has recently shifted to only sponsoring engineers (SOC 15-1252 etc.), not managers.	PM who passes the H-1B filter and invests heavily in a company that will not actually sponsor their role category.	US ETLCA data is historical. A company that sponsored SOC 11-3021 in 2022 may have no intent to do so in 2026. No script currently tracks this shift.
Cognitive demand score treats SOC codes as homogeneous — 'Computer and IT Managers' covers both a PM-equivalent TPM and an engineering VP.	Anyone using the score as a precise filter rather than a directional one.	The score is an average across all incumbents. The variance within SOC 11-3021 is wide enough that the mean is not a reliable individual-role signal.

8. Worked Example — Together AI TPM Role

I ran this mode against a real role I'm currently interviewing for: **Technical Program Manager, GPU Fleet Operations** at Together AI. The scenario is not hypothetical — I have a first-round scheduled.

Inputs used:

- Company: Together AI
- Role SOC estimate: 11-3021 (Computer and Information Systems Managers)
- JD source: Together AI careers page (fetched 2025-06-01)
- OPT timeline: MS graduation August 2026, STEM OPT available

Step-by-step output:

Step 1 — H-1B + Funding	[VERIFIED]
-------------------------	------------

Together AI: Form D filing found, ~14 months ago. H-1B sponsorship record in SOC 15-1299 (Software Developers) confirmed — no record in SOC 11-3021 specifically. FLAG: company sponsors engineers but no direct evidence of sponsoring PM/TPM track. Decision: proceed with caution, not a drop.

Step 2 — Liveness	[VERIFIED]
-------------------	------------

Posting confirmed live as of 2025-06-01 via Playwright. ATS: Ashby. Posting age: ~22 days. Clean signal — this is an active search.

Step 3 — Role Classifier	[INFERRED (manual)]
--------------------------	---------------------

JD keyword count: on-call (2x), SLO (3x), incident command (1x), MTTR (1x), GPU fleet (4x), capacity planning (2x). PM-adjacent signals: roadmap (1x), cross-functional (1x). Manual classification: INFRA_HEAVY. This is consistent with what I found in the actual interview prep — the role is closer to a reliability engineering lead than a PM.

Step 4 — Cognitive Score

[VERIFIED]

SOC 11-3021: verification index 0.81, system judgment 0.74, causal reasoning 0.69. Directionally, this confirms the cognitive premium is present — but the within-SOC variance warning applies. This role's infra focus likely sits at the lower end of PM-equivalence for that cognitive profile.

Step 5 — Decision

[OUTPUT]

STRETCH. Not SKIP because: funding is recent, posting is live, company does sponsor, and cognitive scores confirm transferable premium. Not APPLY because: no direct SOC 11-3021 sponsorship history, and INFRA_HEAVY classification means interview prep needs to address the engineering gap proactively.

What I would change based on seeing this output:

1. Build `tpm_classifier.py`. The manual keyword count took ~15 minutes and is the single biggest bottleneck in the workflow. The logic is simple enough that even a regex-based v1 would cut that to under a minute.
2. Add a SOC sponsorship specificity check — not just 'does this company sponsor' but 'has this company sponsored 11-3021 specifically.' The current H-1B data has this field; the query script just isn't filtering on it precisely enough.
3. The STRETCH flag needs a next-action attached to it — right now it's a dead end. A useful output would be: 'STRETCH — recommend recruiter screen to confirm role-type before full prep investment.' That's a workflow addition, not a data gap.